

Use Case 1

Real-Time Pipeline Intelligence & Partnership Opportunity Scanning

Software Requirement Specification & Solution Document

[Architecture](#) · [Sequence & Collaboration Diagrams](#) · [Implementation Reference](#) · [Live Results](#)

Field	Value
Document	SRS & Solution — Use Case 1
Status	Implemented, deployed and verified
Generated	2026-08-09 — from live system data
Service	eugene_scout v1.0.0
Data sources	ClinicalTrials.gov · Europe PMC · USPTO ODP · SEC EDGAR
Datastore	s3://eugene-scout-087084717211 (staging + curated)
Verification	374 unit/integration tests · 46 browser tests · live scan
Audience	BD leadership, solution architects, engineering

Every figure, company name and result in this document is read from the running system at generation time. Nothing is illustrative and nothing is hand-typed — if the live data were unavailable, this document would not have been produced.

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Part A — Executive summary

CSL's Business Development group identifies partnership targets through periodic analyst reviews, conference attendance and relationship networks. That approach is episodic: a biotech advancing into Phase I/II, or raising a financing round, can go unnoticed until a competitor has already opened discussions. The monitorable surface — ClinicalTrials.gov, the literature, patent filings and SEC disclosures — is far larger than any team can read.

Use Case 1 asked for an autonomous scanner that watches those sources continuously, ranks what it finds against CSL's strategic filter, and delivers a weekly briefing of partnership candidates. **That system is built, deployed and running unattended on a daily schedule.**

Measure	Value
Signals under management	122
Companies ranked	33
Therapeutic areas scanned	5
Signals in the last 7 days	16
Partners identified this week	6
Scan duration (all areas, 4 sources)	~80s
Sources operational	4 of 4
Automated tests passing	374 unit/integration + 46 browser

The design decision that matters most

Ranking is produced by a deterministic weighted function of six named factors — not by a language model. A model asked to rank twice answers differently and, asked why, invents a reason. The LLM is confined to writing one paragraph of prose *about a decision already made numerically*, and any sentence it produces containing a number absent from its input is discarded automatically. This is what makes the output challengeable by a BD leader rather than merely presentable.

Part B — Use case definition

The source specification is restated below as numbered, testable requirements. Each carries its implementation location and the verification that demonstrates it. 'Verified by' names a real test or a live observation, not an intention.

ID	Requirement	Implemented in	Verified by
FR-1	Scheduled orchestrator triggers the scan on a defined schedule	api.py — APScheduler CronTrigger, 02:00 UTC daily	Cron run observed in run history
FR-2	Sub-agents ingest ClinicalTrials.gov, literature, patents (USPTO/EPO), SEC EDGAR	collectors/{trials,literature,patents,edgar,epo}.py	38 contract tests; live scan
FR-3	Entities extracted and linked to CSL's internal ontology	enrich.py + graph.py → Neo4j via eugene_ws	22 tests; live graph annotations
FR-4	Signals ranked against configurable BD criteria	scoring.py — 6 weighted factors	22 determinism//bounds tests
FR-5	Ranked output written to a structured store	store.py → S3 curated/ tier	31 storage tests
FR-6	Report-generation agent produces a formatted briefing	digest.py → /briefing → artifact export	5 browser tests
FR-7	High-scoring signals pushed with a one-paragraph rationale	notify.py — webhook/SMTP, off by default	16 tests; live webhook verified
FR-8	8–12 ranked companies with rationale, sources and next action	digest.py::build/_next_action	Browser test asserts all four present
NFR-1	Human-in-the-loop: flag low confidence, never suppress it	review.py — 4 structural conditions	Flagged signals still queryable
NFR-2	Auditability: every action and source traceable	runs/ manifests + score_breakdown + signal_sources	Every signal links to its source
NFR-3	Configurable thresholds managed as configuration, not code	config.py + S3 scan_config.json + Settings UI	Config change persists across reload
NFR-4	Security: least privilege, authenticated access	security.py + scoped IAM policy	30 auth tests

Out of scope, and why

Item	Reason
Evaluate Pharma, Citeline	Commercial subscriptions; no entitlement available
EPO OPS patent source	Implemented and unit-tested, ships disabled — OAuth key required to verify the live contract
AWS deployment	Runs on Docker Compose; Terraform deferred by agreement
SMTP notification	Implemented, untested — webhook delivery was verified live instead

Part C — Results: what the system actually found

The following is live output, read from the running scanner on 2026-08-09. The window is the seven days to 2026-08-09 (publications dated on or after 2026-08-02).

C.1 Partners identified in the last seven days

#	Company	Score	Signal type	Published	Evidence
1	Attovia Therapeutics, Inc.	67.2	Corporate	2026-08-04	Attovia Therapeutics, Inc. filed S-1/A
2	Octapharma	65.3	Trial	2026-08-06	Study of IV Human Plasma-derived C1 Esterase Inhibitor Con...
3	CSL Behring	63.2	Trial	2026-08-05	Phase 3, Open-label, Single-dose Study of CSL222 in Adoles...
4	KalVista Pharmaceuticals, Ltd.	62.3	Trial	2026-08-05	Open-Label Safety, PK, and Efficacy Trial of Sebetralstat ...
5	Ionis Pharmaceuticals, Inc.	61.3	Trial	2026-08-05	Donidalorsen Treatment in Children With Hereditary Angioed...
6	Beam Therapeutics Inc.	51.3	Patent	2026-08-06	METHODS AND COMPOSITIONS FOR TREATING ALPHA-1 ANTITRYPSIN ...

Attribution is deliberately conservative: a trial sponsored by a university or government body is kept as a signal but carries *no* company attribution, because an institution is not a partner CSL can transact with. That is why this list is shorter than the signal count.

C.2 Patents (1 in the window)

Score	Published	Title	Company / source
51.3	2026-08-06	METHODS AND COMPOSITIONS FOR TREATING ALPHA-1 ANTITRYPSIN DEFICIENCY	Beam Therapeutics Inc.

C.3 Publications (5 in the window)

Score	Published	Title	Company / source
83.6	2026-08-04	[Current and future therapies for bradykinin-mediated angioedema].	europepmc_lit
74.7	2026-08-07	Safety and tolerability of a new intravenous immunoglobulin 10% in patient...	europepmc_lit
58.0	2026-08-06	Impaired immune response to polysaccharide antigens and warning signs in c...	europepmc_lit
57.0	2026-08-07	p62/SQSTM1-KEAP1 complex prevents clearance of ubiquitinated Z alpha-1 ant...	europepmc_lit
45.9	2026-08-06	Quality of Life in People With Haemophilia: Results in the Framework of a ...	europepmc_lit

C.4 Clinical trials (9 in the window)

Score	Published	Title	Company / source
76.0	2026-08-07	Immunotherapy for Neurological Post-Acute Sequelae of SARS-CoV-2	clinicaltrials
67.7	2026-08-07	BCMA/GPRC5D Trispecific Antibody Treatment for Newly Diagnosed Amyloidosis...	clinicaltrials
65.3	2026-08-06	Study of IV Human Plasma-derived C1 Esterase Inhibitor Concentrate in Pati...	Octapharma
63.2	2026-08-05	Phase 3, Open-label, Single-dose Study of CSL222 in Adolescent Male Subjec...	CSL Behring
62.3	2026-08-05	Open-Label Safety, PK, and Efficacy Trial of Sebetralstat (KVD900) in Pedi...	KalVista Pharmaceuticals, Ltd.
61.3	2026-08-05	Donidalorsen Treatment in Children With Hereditary Angioedema	Ionis Pharmaceuticals, Inc.
57.5	2026-08-05	A Phase 2 Study of Foscenvivint in Liver Cirrhosis Patients Caused by HIV/...	clinicaltrials
56.3	2026-08-05	Donidalorsen Expanded Access Program for Patients With Hereditary Angioede...	Ionis Pharmaceuticals, Inc.
44.5	2026-08-06	Rapid or Ultra-rapid Diagnosis of Myocardial Infarction Using High-Sensiti...	clinicaltrials

C.5 Corporate filings (1 in the window)

Score	Published	Title	Company / source
67.2	2026-08-04	Attovia Therapeutics, Inc. filed S-1/A	Attovia Therapeutics, Inc.

C.6 Ranked partner watchlist

Company scores are derived entirely from the signals attributed to them — there is no separately maintained score that could disagree with the evidence.

Rank	Company	Score	Signals	High	Areas
1	Regeneron Pharmaceuticals, Inc.	72.5	3	0	immunoglobulin
2	Attovia Therapeutics, Inc.	69.2	2	0	immunoglobulin
3	KALVISTA PHARMACEUTICALS LIMITED	68.3	4	0	hereditary_angioedema
4	Pharvaris Netherlands B.V.	67.1	2	0	hereditary_angioedema
5	Gilead Sciences, Inc.	66.0	1	0	immunoglobulin
6	Octapharma	65.3	1	0	hereditary_angioedema
7	Bioverativ, a Sanofi company	64.6	2	0	hemophilia
8	Hoffmann-La Roche	64.6	1	0	hemophilia
9	Astria Therapeutics, Inc.	64.3	3	0	hereditary_angioedema
10	Intellia Therapeutics, Inc.	63.9	3	0	hereditary_angioedema
11	Ionis Pharmaceuticals, Inc.	63.3	2	0	hereditary_angioedema
12	CSL Behring	63.2	3	0	alpha1_antitrypsin, hemophilia, he
13	Sanofi	62.9	2	0	immunoglobulin
14	AIRNA Corporation	62.3	1	0	alpha1_antitrypsin
15	iOmx Therapeutics AG	61.4	1	0	immunoglobulin
16	Charité - Universitaetsmedizin Berlin	61.4	1	0	immunoglobulin
17	Biomolecular Holdings LLC	60.9	1	0	immunoglobulin
18	Takeda Pharmaceutical Company Limited	60.8	3	0	hereditary_angioedema

C.7 Corpus composition and data currency

Signal type	Count		Priority band	Count
Publication	59		high	2
Trial	25		med	54
Patent	33		watch	66
Corporate	5			

Source	Newest record	Age (days)	Signals
clinicaltrials	2026-08-07	2	25
europemc_lit	2026-08-07	2	59
europemc_pat	2026-08-06	3	33
sec_edgar	2026-08-04	5	5

Per-source dates are extracted from the upstream records themselves, not from when the scan ran. The distinction matters: a panel can truthfully report 'scanned 20 minutes ago' while its newest patent is two months old, and only the second number indicates staleness.

Part D — Solution architecture

All diagrams in this section are generated from Mermaid source held under `docs/diagrams/*.mmd` and rendered at build time. The source is reviewable text under version control, so a diagram is changed by editing a line rather than by recreating an image.

D.1 Component architecture — as deployed today

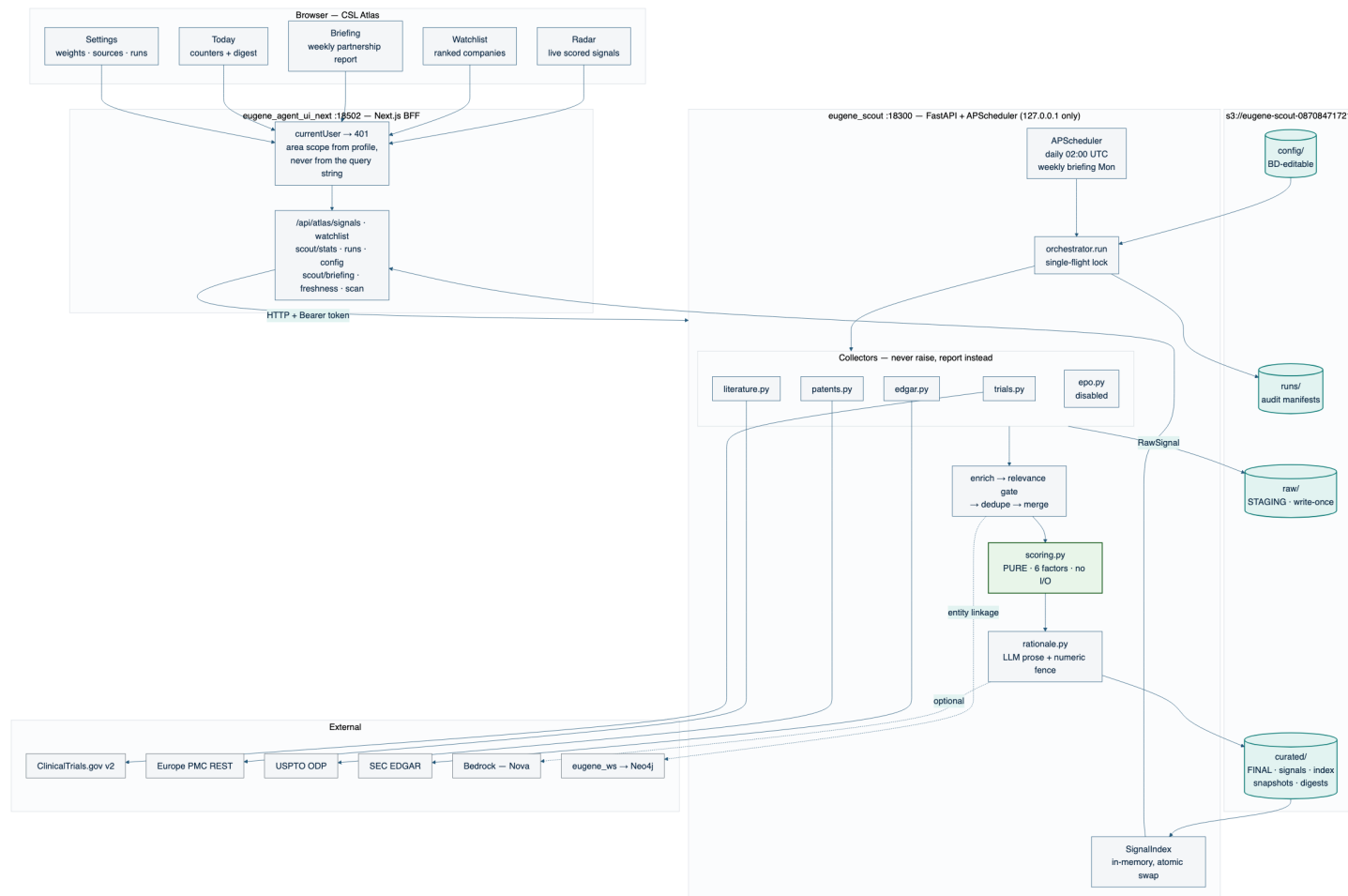


Figure 1 — Component architecture on Docker Compose. The browser talks only to the Next.js BFF; the BFF is the only thing that holds the scanner's bearer token.

Why S3 is the only database

The corpus is a few thousand signals, read-mostly, and rebuilt wholesale by each scan — the shape object storage serves well and a relational store buys nothing for. Ad-hoc filtering, the one thing Postgres would have provided, comes instead from an in-memory index rehydrated from a single curated object per area. Measured query latency stays sub-second at 50,000 signals, an order of magnitude beyond current volume.

D.2 System sequence — configuration through to report

One scheduled scan end to end. Three orderings here are deliberate rather than incidental: staging is written *before* any judgement is applied, the dated snapshot is written *before* the live objects, and the in-memory index is refreshed *before* notifications are dispatched.

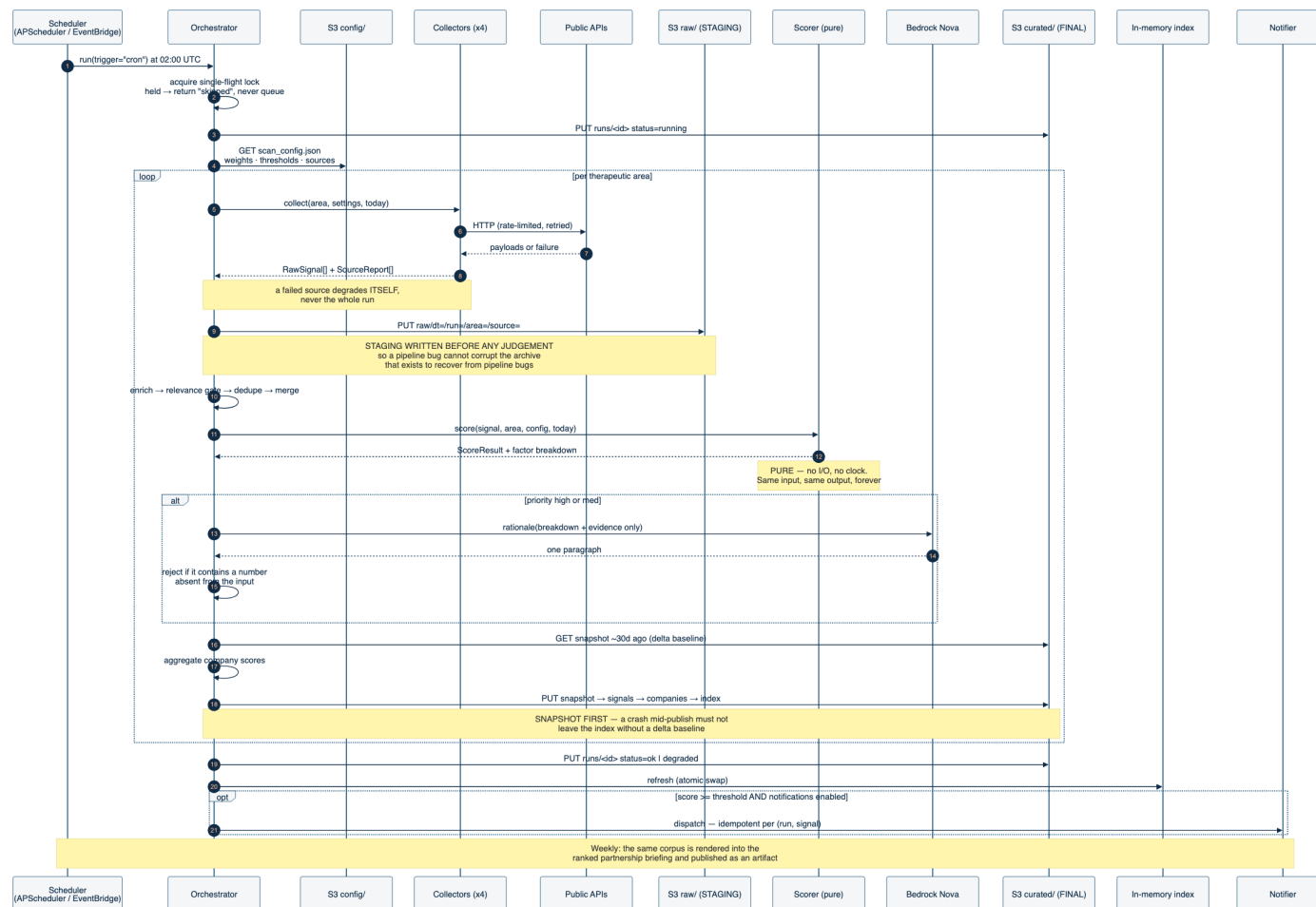


Figure 2 — System sequence diagram for one scheduled scan.

D.3 Collaboration diagram

Where the sequence diagram shows ordering over time, this shows which object is responsible for asking whom. Messages are numbered in execution order.

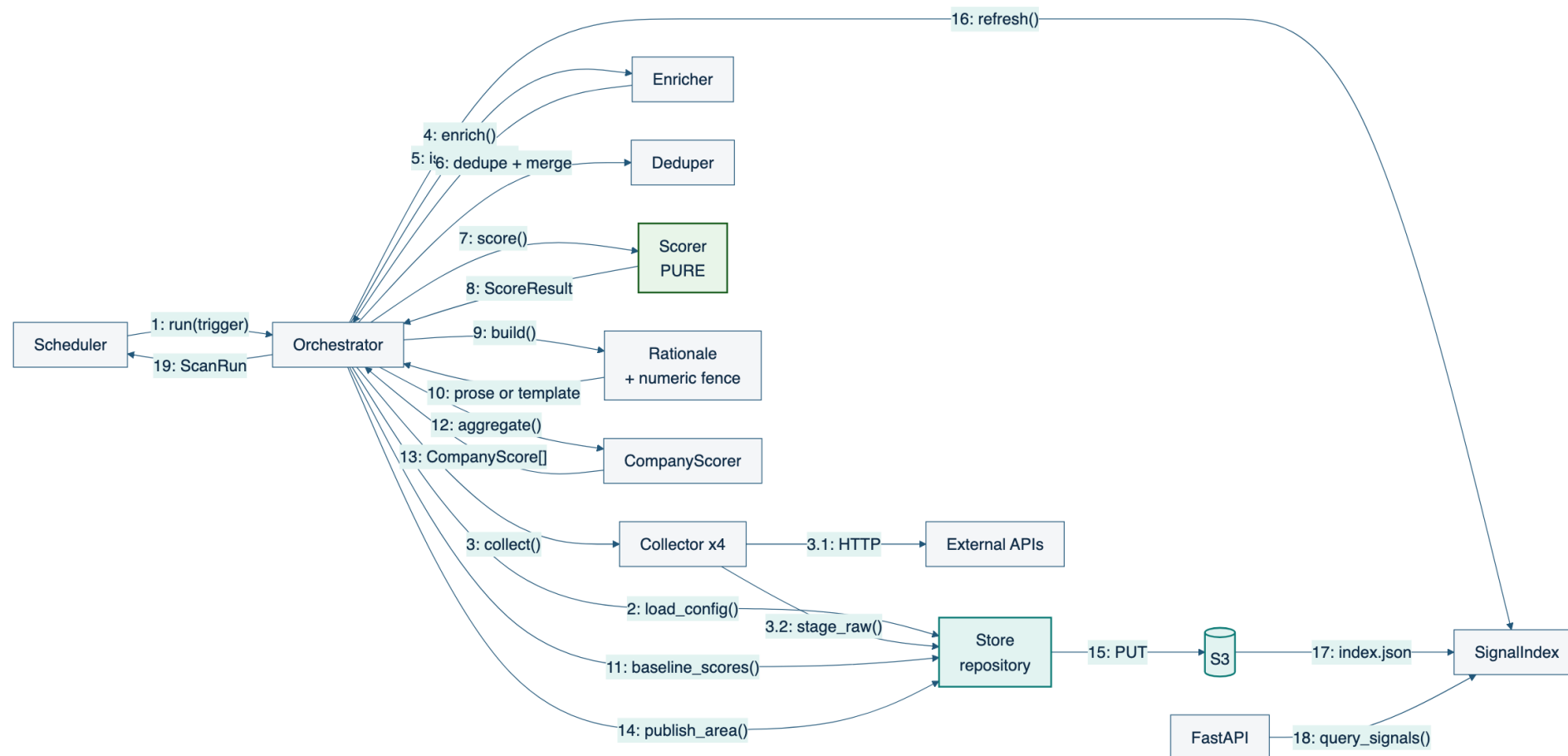


Figure 3 — Collaboration diagram. The Scorer has no outbound edges: it is a pure function whose every input arrives as an argument, which is what makes a rank reproducible and challengeable.

D.4 Configuration to report — the full chain

What a BD analyst controls, and how a change to it reaches the weekly briefing. The chain is shown in two halves: configuration and collection here, analysis and delivery overleaf.

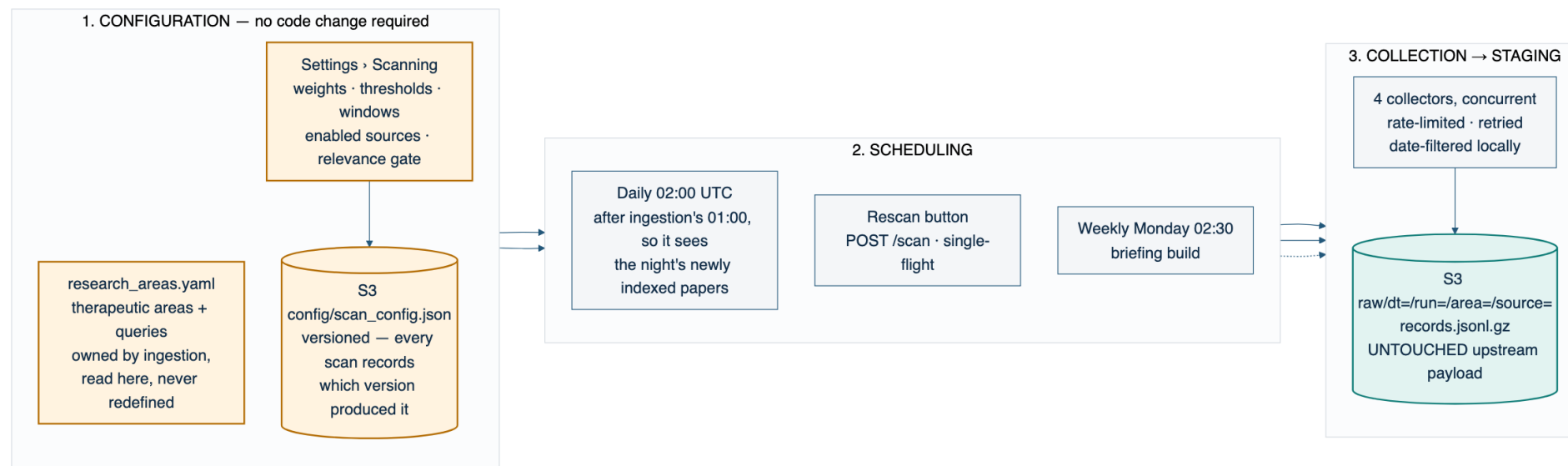


Figure 4a — Configuration and scheduling through to staged records. Nothing in the amber boxes requires a code change or a deployment.

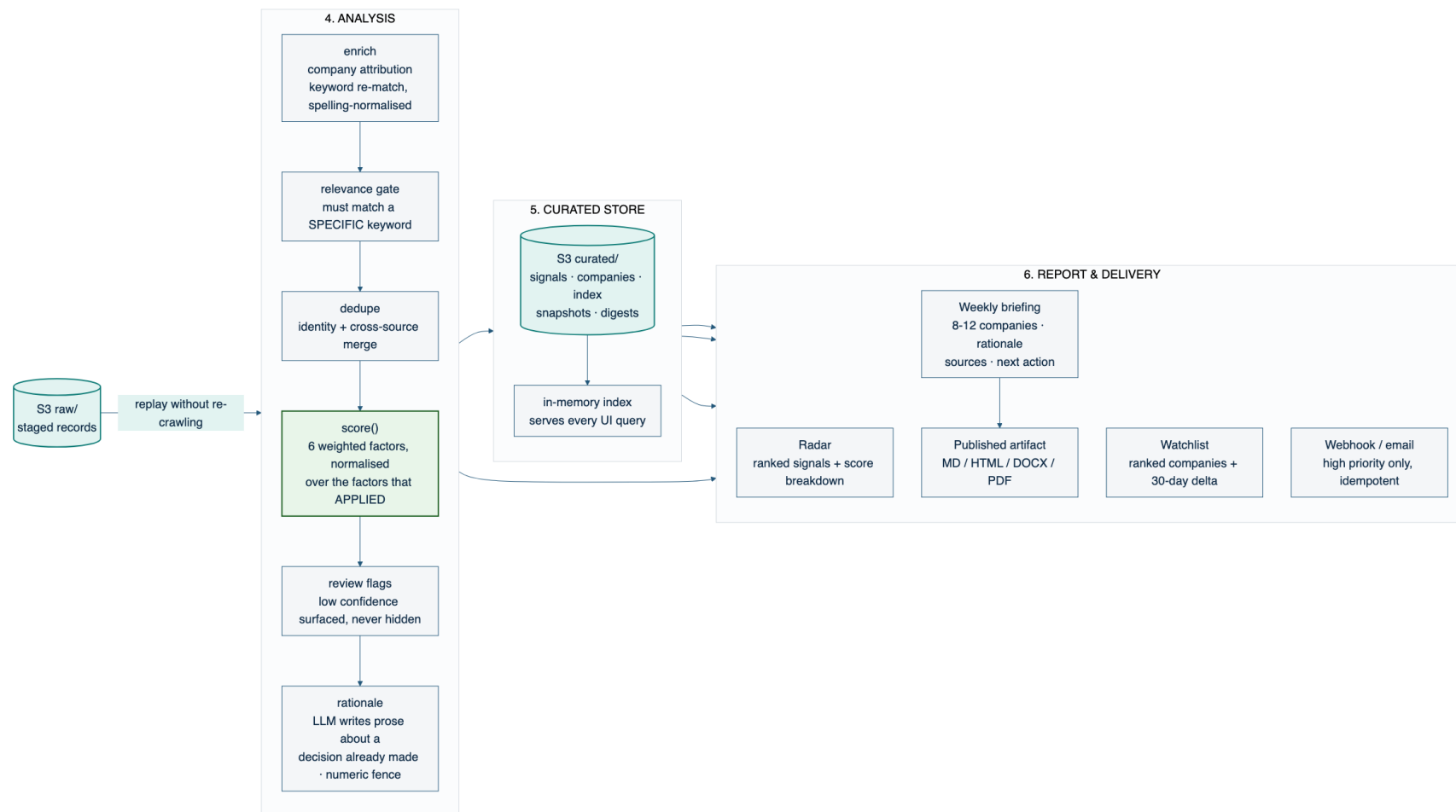


Figure 4b — Staged records through analysis and the curated store to the delivered report. The dashed edge from raw/ is the replay path: a scoring change can be re-applied over history without re-crawling any public API.

D.5 Where the data lives

Two tiers with deliberately different guarantees. **raw/** is the staging tier: untouched upstream payloads, written once, never mutated, expiring at 90 days. It exists so a scoring change can be replayed over history without re-crawling four public APIs, and so the question 'what did we actually see on the 3rd?' has an answer months later. **curated/** is the final tier: scored signals, company scores, the per-area index the API serves from, immutable dated snapshots that provide the 30-day delta baseline, and the weekly briefings.

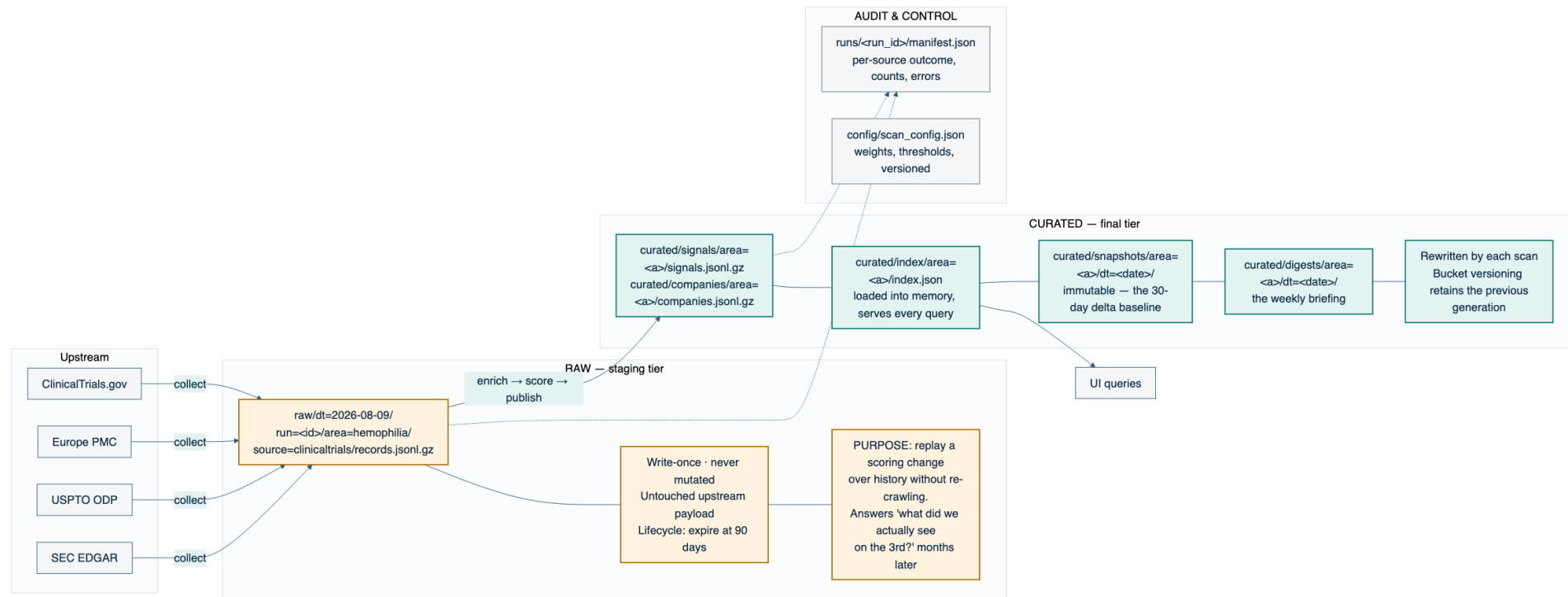


Figure 5 — Raw (staging) and curated (final) tiers, with the audit and control objects alongside.

Concrete object paths in the live bucket

s3://eugene-scout-087084717211/

```
raw/dt=2026-08-09/run=20260809T020000Z-a1b2c3d4/
  area=hemophilia/source=clinicaltrials/records.jsonl.gz      STAGING

curated/signals/area=hemophilia/signals.jsonl.gz             FINAL
curated/companies/area=hemophilia/companies.jsonl.gz
curated/index/area=hemophilia/index.json                     <- read into memory at boot
curated/snapshots/area=hemophilia/dt=2026-08-09/signals.jsonl.gz
curated/digests/area=hemophilia/dt=2026-08-09/digest.json

runs/20260809T020000Z-a1b2c3d4/manifest.json                 <- per-source audit record
config/scan_config.json                                       <- weights, thresholds (v27)
```

Part D (ii) — Production architecture on AWS

The service runs today on Docker Compose. This section describes the target AWS deployment and, more usefully, **how little of it is a rewrite**. The application code is unchanged; what moves is the scheduler, the credential mechanism, and the edge.

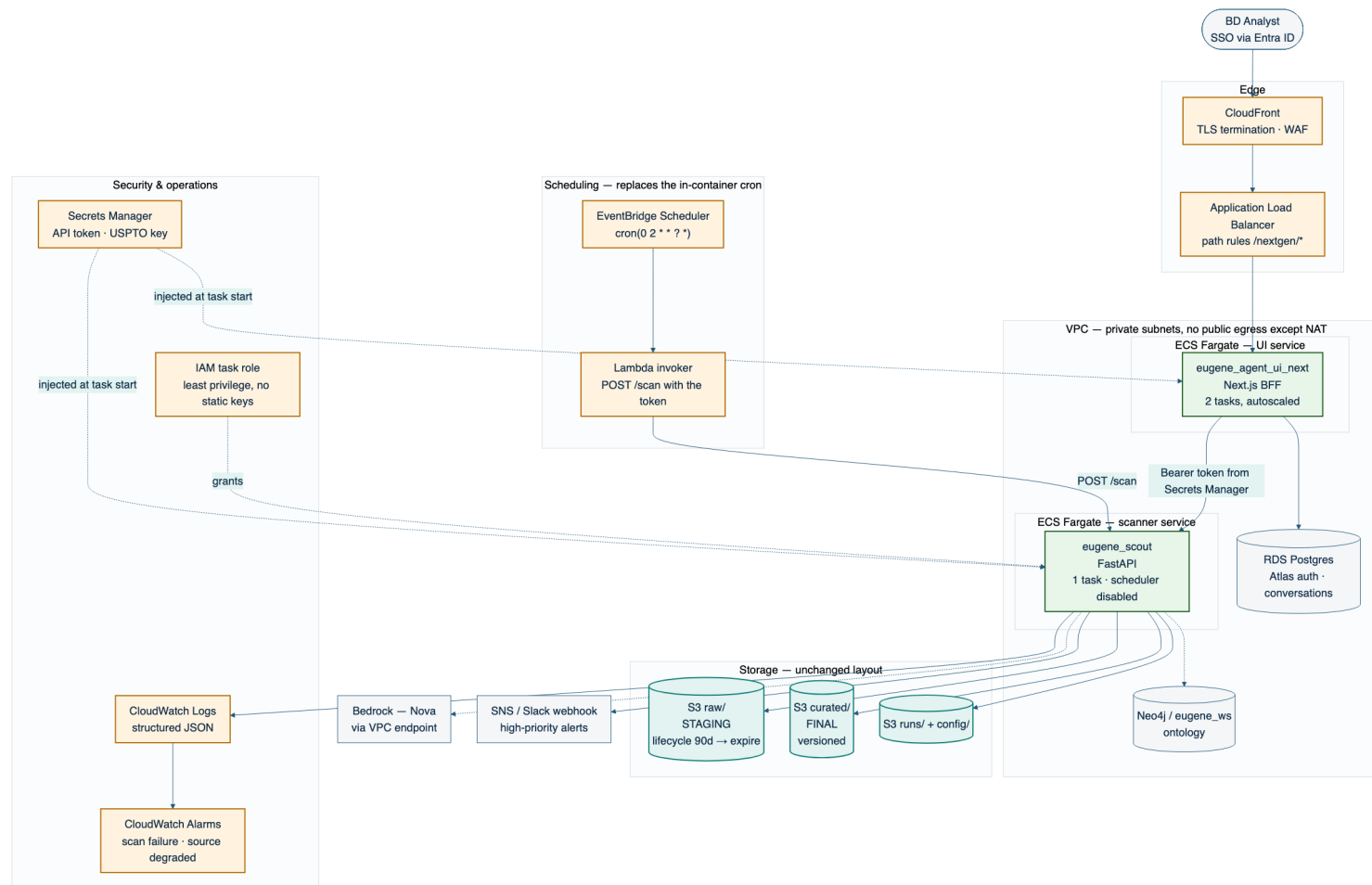


Figure 6 — Target AWS architecture. Amber components are the only ones that change; green are the same containers running unmodified; teal is storage, whose layout does not change at all.

D.6 Local today vs AWS target

Concern	Local today	On AWS	Code change?
Compute	Docker Compose on one host	ECS Fargate — UI service (2 tasks, autoscaled) and scanner service (1 task)	None
Scheduling	APScheduler cron inside the container	EventBridge Scheduler → Lambda → POST /scan	None — /scan already exists and is authenticated
Object storage	S3 (already)	S3 — same bucket, same key layout, same lifecycle	None
Credentials	Static access keys in docker.env	IAM task role; boto3's default chain picks it up	None — config.s3_credentials() already falls through
Secrets	gitignored docker.env	Secrets Manager, injected at task start	None — same env var names
TLS	Plaintext inside the compose network	CloudFront + ALB terminate TLS; VPC-internal hop	None
Ingress	Published localhost ports	ALB path rules; scanner has NO listener outside the VPC	None
Logging	docker logs (structured JSON)	CloudWatch Logs — the JSON is already query-shaped	None
Alerting	Webhook, off by default	SNS → Slack/email, plus CloudWatch alarms on scan failure	Configuration only
Auth to scanner	Bearer token from docker.env	Same token, sourced from Secrets Manager	None
Relational data	atlas_postgres container	RDS Postgres, Multi-AZ	Connection string only

Why the scheduler moves out of the container

In-container APScheduler is correct for a single host: it ships with the stack and survives a rebuild. On ECS it becomes a liability — scale the service to two tasks and both fire the same scan, issuing duplicate requests to rate-limited public APIs and racing each other on the same S3 keys. EventBridge invoking POST /scan makes the trigger external and singular, and the endpoint it calls already exists, is authenticated, and is single-flight guarded. Set SCOUT_SCHEDULER_ENABLED=false on the task and nothing else changes.

D.7 Migration sequence

Ordered so each step is independently verifiable and reversible.

#	Step	Detail	Verified when
1	Provision the scoped IAM role	infrastructure/iam/ policy already written; attach to the ECS task role instead of an IAM user	verify-scout-iam.sh reports 10/10
2	Move secrets to Secrets Manager	Same variable names, so the container reads them identically	/health reports auth: enabled
3	Push images to ECR	Both Dockerfiles build unchanged	Image digests recorded
4	Stand up ECS services behind an ALB	UI public via CloudFront; scanner private, no public listener	UI loads; scanner unreachable from outside the VPC
5	Disable the in-container scheduler	SCOUT_SCHEDULER_ENABLED=false	No cron entries in the task log
6	Create the EventBridge schedule + Lambda invoker	cron(0 2 * * ? *) UTC	A run appears with trigger=cron
7	Wire CloudWatch alarms	Alarm on scan status=failed and on any source degraded two runs running	Alarm fires on an induced failure
8	Cut DNS over	Retain the local stack until a full week of clean runs	Seven consecutive successful scheduled runs

D.8 What stays exactly the same

- **The S3 layout.** Same bucket, same raw/ and curated/ prefixes, same Hive-style partitioning. The data written locally today is directly readable by the AWS deployment.
- **Every line of scanner and UI application code.** The credential chain, the scheduler toggle and the secret source are all existing configuration points.
- **The scoring function and its stored configuration.** scan_config.json moves with the bucket, so weights tuned locally carry over with their version history.
- **The test suite.** It is hermetic — no network, in-process S3 — so it runs identically in CI.

One thing that must change before production traffic

The UI-to-scanner hop is plaintext HTTP. On one host inside a compose network that is acceptable; spanning ECS tasks it is not. Either terminate TLS at an internal ALB or enable service-to-service encryption. This is the single item that is a genuine gap rather than a configuration move.

Part E — Implementation reference

Every file in the service, what it is responsible for, and the design constraint it enforces. This is the section a developer joining the project should read first.

E.1 Scanner service — `agents/eugene-scout/src/scout/`

File	Responsibility	LoC
<code>api.py</code>	HTTP surface + APScheduler. Two cron jobs (daily scan 02:00 UTC, weekly briefing Monday). Single-flight asyncio.Lock; blocking work in a threadpool so the event loop keeps serving.	223
<code>orchestrator.py</code>	The pipeline. collect → stage → enrich → gate → dedupe → score → rationale → aggregate → publish. Owns failure isolation: a source degrades itself, an area degrades itself.	159
<code>scoring.py</code>	PURE deterministic scoring. Six factors, normalised over the factors that applied. No I/O, no clock — `today` is an argument so historical replay is meaningful.	75
<code>rationale.py</code>	The only LLM call, fenced. Generates prose about an existing score and discards any output containing a digit absent from its input.	142
<code>review.py</code>	Human-in-the-loop confidence flagging. Four structural conditions; flags, never suppresses.	29
<code>dedupe.py</code>	Identity dedup (source:external_id) + conservative cross-source merge. Compares only across sources — the optimisation that took 3,000 records from 14.6s to under a second.	93
<code>enrich.py</code>	Company attribution and keyword re-matching. Drops non-commercial sponsors: a university is not a partner CSL can transact with.	48
<code>companies.py</code>	Watchlist aggregation. Best signal + bounded support bonus, so extra evidence can never lower a company's rank.	63
<code>graph.py</code>	Knowledge-graph entity linkage via eugene_ws. Context only — deliberately NOT a scoring input.	60
<code>store.py</code>	Repository. The only module that maps domain objects onto S3 keys.	135
<code>storage.py</code>	S3 primitives and the single source of truth for the key layout.	134
<code>index.py</code>	In-memory query index, atomically swapped after each scan. This is what replaces the database.	146
<code>models.py</code>	Pydantic contracts for every boundary. Validators reject malformed upstream data at the edge.	215
<code>config.py</code>	Two config tiers: environment (deploy) and S3-backed scan settings (BD-editable).	111
<code>areas.py</code>	Loads the therapeutic taxonomy from the ingestion YAML. Never redefines it.	82
<code>text.py</code>	Biomedical normalisation. Exists because 'Haemophilia' matched no keywords on a live run.	58
<code>digest.py</code>	The weekly briefing: 8–12 ranked companies, rationale, sources, next action.	76
<code>notify.py</code>	Webhook/SMTP dispatch. Off by default, idempotent per (run, signal).	107
<code>security.py</code>	Bearer-token auth applied app-level so a new route is protected by default.	—
<code>logging_setup.py</code>	Structured JSON logging with X-Request-Id correlation.	—
<code>collectors/base.py</code>	Collector protocol, rate limiting, retry, and the never-raise contract.	93

File	Responsibility	LoC
collectors/trials.py	ClinicalTrials.gov v2. Industry sponsors only for attribution.	49
collectors/literature.py	Europe PMC. Pushes the date window upstream; requests abstracts.	42
collectors/patents.py	USPTO Open Data Portal. Replaced Europe PMC's patent index, which stops in 2012.	64
collectors/edgar.py	SEC EDGAR full-text, filtered to biotech SIC codes.	62
collectors/epo.py	EPO OPS. Implemented, unit-tested, ships disabled pending an OAuth key.	133

E.2 User interface — agents/eugene-agent-ui-next/

File	Responsibility
components/atlas/views/RadarView.tsx	Live signal list, filters, score breakdown, dismiss, rescan
components/atlas/views/WatchlistView.tsx	Ranked company table with 30-day deltas
components/atlas/views/BriefingView.tsx	Weekly briefing + publish-as-artifact
components/atlas/DataFreshness.tsx	The 'As of ' header chip and per-source currency panel
components/atlas/ScanningSettings.tsx	Weights, thresholds, sources, run history
components/atlas/SinceYouLastLooked.tsx	Real 'new since last visit' counters
components/atlas/NeedsAttention.tsx	Top-priority signals on the Today view
lib/atlas/scout.ts	Types mirroring the pydantic models field-for-field
lib/atlas/scoutClient.ts	Server-side client. Never throws — returns degraded payloads
hooks/useScout.ts	Client fetching with request sequencing so filters cannot race
app/api/atlas/signals/route.ts	BFF: auth gate, server-set area scope
app/api/atlas/scout/*/route.ts	BFF: stats, runs, config, scan, briefing, freshness, notify-test

E.3 Data layout in S3

s3://eugene-scout-087084717211/

```

raw/dt=<date>/run=<id>/area=<area>/source=<src>/records.jsonl.gz    STAGING
| untouched upstream payloads, write-once, 90-day lifecycle
| written BEFORE enrichment, so a pipeline bug cannot corrupt the
| archive that exists to recover from pipeline bugs

curated/signals/area=<area>/signals.jsonl.gz                      FINAL
curated/companies/area=<area>/companies.jsonl.gz
curated/index/area=<area>/index.json    <- loaded to memory, serves queries
curated/snapshots/area=<area>/dt=<date>/ <- immutable; 30-day delta baseline
curated/digests/area=<area>/dt=<date>/   <- weekly briefing

runs/<run_id>/manifest.json    <- audit trail, per-source outcomes
config/scan_config.json      <- BD-editable weights & thresholds

Hive-style partitioning throughout, so Athena / Glue remain available
later without a migration.
```

Figure 5 — Hive-style partitioning throughout, so Athena/Glue remain available without migration.

Part F — Scoring and explainability

The specification asks for recommendations BD leaders can 'interrogate and challenge'. That requirement is what forced a deterministic scorer.

$$\text{score} = 100 \times \Sigma(\text{weight}_i \times \text{factor}_i) \div \Sigma(\text{weight of factors that APPLIED})$$

Normalising by the *applied* weight rather than the configured total is deliberate. A literature hit has no development stage; scoring that factor zero would systematically punish an entire source for lacking a property it cannot have.

Factor	Weight	Measures	Design note
area_fit	0.30	Keyword overlap with the declared taxonomy	Saturating — length is not relevance
stage_fit	0.20	Development stage	Peaks at Phase I/II, where a deal is still available
recency	0.15	Age against that source's own window	Per-source, so patents stay competitive
modality_fit	0.15	Overlap with CSL capability	Separates actionable from merely interesting
corroboration	0.10	Independent sources for one event	Strongest evidence the system produces alone
company_context	0.10	Prior activity from this company	Small — the point is finding new names

Priority bands: **high** ≥ 75.0, **med** ≥ 55.0, otherwise watch. All weights, thresholds, source windows and the relevance gate live in S3 and are editable from Settings — configuration, not code. Current config version: **v27**.

The anti-hallucination fence

The LLM never produces a score, rank, date or company name. It receives an already-computed factor breakdown and writes one paragraph. Every digit in its output is then checked against its input, and any sentence containing an unsupported number causes the whole paragraph to be discarded in favour of a deterministic template. On the live corpus this fires: rationale provenance is labelled AI-WRITTEN or COMPUTED in the UI, and rejected outputs are additionally flagged for human review.

F.1 The relevance gate

A record must match at least one *specific* (non-generic) keyword to enter the curated tier. Every term on the generic list is there because of a measured false positive:

Generic term	False positive it admitted
gene therapy	USPTO patents for hearing loss and phenylketonuria entered the hemophilia area
inhibitor	An oncology checkpoint-inhibitor trial entered the hemophilia area
augmentation	Patents from Palo Alto Networks ('prompt augmentation') and OPTUM ('authorization request augmentation') entered the alpha-1 antitrypsin area — found while compiling this document

Part G — Quality and verification

Layer	Tests	Runtime	What it protects
Unit — pure logic	~150	<1s	Scoring determinism, text normalisation, dedup, aggregation
Contract — collectors	38	<1s	Parsers match what the APIs actually send (recorded fixtures)
Integration — storage/pipeline	89	~10s	S3 key layout, publish ordering, run identity
API — HTTP surface	40	~5s	Status codes, auth gating, cold-start shapes
Security	30	<1s	Token enforcement, protected-by-default, credential scoping
Stress — scale & hostile input	34	~25s	50k-signal latency, $O(n^2)$ guards, unicode/oversized input
Browser — end-to-end	46	~2m	Real data, working links, degraded states

374 Python tests · 86% line coverage · 46 browser tests. The Python suite is hermetic — no network, in-process S3 via moto, frozen clock. The browser suite runs against real scan output, deliberately, because the defect class this feature exists to prevent is 'the panel shows something plausible that is not true', which a mocked test cannot see.

G.1 Defects found and fixed during development

Each of these was found by running the system, not by reading it. All are now regression-tested.

Defect	Impact	Fix
Europe PMC patent index abandoned	139,427 records for 2011–12 and zero after; no filtering can make a frozen corpus fresh	Replaced with USPTO ODP
British spellings matched nothing	'Haemophilia' matched no keywords — every European journal silently sank	Biomedical normalisation
Lucene operator precedence	AND binds tighter than OR, so the date filter applied to only the last keyword	Parenthesised the clause
USPTO 404 on empty results	Whole areas marked degraded on quiet nights	Treated as zero results
Company records split	Bioverativ appeared twice, halving its evidence	Ownership/industry suffix stripping
Top-k mean punished coverage	A better-documented company ranked lower	Best signal + bounded bonus
Container-only import crash	parents[4] valid in a checkout, IndexError at /app — crash-looped on first deploy	Guarded + shallow-path test
Run-id collision	Second-precision ids overwrote each other's manifest	8-hex suffix
O(n ²) dedup	14.6s for 3,000 records	Cross-source-only comparison → <1s
Login hydration race	Controlled inputs wiped browser autofill	Adopt pre-hydration DOM values
Unauthenticated API	Config rewritten from an unauthenticated curl against the live container	Bearer token + localhost bind
Compose env override	environment: silently blanked env_file: values	Precedence moved into code
Generic keyword false positives	Networking/insurance patents in a biology area	Evidence-driven generic list

Part H — Operations and security

H.1 Deployment

```
docker compose up -d eugene_scout eugene_agent_ui_next
```

```

UI          http://localhost:18502/nextgen      Radar ■4 · Watchlist ■5 · Briefing ■6
Scanner     http://localhost:18300             127.0.0.1 only, bearer token
Schedule    daily scan 02:00 UTC · weekly briefing Monday 02:30 UTC
Limits      mem 1g · cpus 2.0
```

H.2 Security posture

Control	Implementation
API authentication	Bearer token on every endpoint except /health, applied app-level so a new route inherits it
Network exposure	Port bound to 127.0.0.1; the UI reaches the scanner over the compose network
API documentation	/docs, /redoc, /openapi.json disabled whenever auth is on
Credential scope	Least-privilege IAM policy limited to the scout bucket (provisioning script supplied)
Secret handling	Tokens and keys in gitignored docker.env; verified absent from every commit
Token comparison	Constant-time — a short-circuiting check leaks the prefix to a timing attack
Notifications	Off by default; a docker compose up cannot message the BD team
Audit trail	Per-run manifests with per-source outcomes; every signal traces to a source URL

H.3 Runbook

```

# Is it healthy?
curl -s localhost:18300/health | python3 -m json.tool

# Did the scheduled scan run? (`trigger: cron` is the proof; S3 outlives container logs)
curl -s -H "Authorization: Bearer $TOKEN" "localhost:18300/runs?limit=10"

# Which source has gone quiet?
curl -s -H "Authorization: Bearer $TOKEN" localhost:18300/freshness

# Force a scan
curl -X POST -H "Authorization: Bearer $TOKEN" localhost:18300/scan

# Logs (JSON; filter by request id, area or level)
docker logs eugene-scout --tail 100
```

An empty Radar is not a broken Radar

If the scanner is unreachable the UI shows an explicit 'Scanning service unavailable' banner rather than a blank list. Distinguishing 'nothing found' from 'nothing loaded' is a deliberate property: a quiet news day and an outage must never look the same to an analyst.

Part I — Conclusion

Use Case 1 is implemented, deployed and verified. The scanner runs unattended on a daily schedule, reads four public sources, and currently manages 122 scored signals across 5 therapeutic areas, ranking 33 companies. In the last seven days it surfaced 16 new signals and 6 commercially-attributed partners, each traceable to the upstream record it came from.

The system's defining property is that its output is challengeable. Every rank decomposes into six named factors with their inputs; every signal links to its source; every scan leaves an audit manifest recording which feeds succeeded and which failed. Where the system is uncertain it says so rather than hiding the row, and where prose was written by a language model it is labelled as such.

I.1 Delivered

- Autonomous scanner across ClinicalTrials.gov, Europe PMC, USPTO and SEC EDGAR, on a daily cron
- Deterministic, configurable, fully explainable ranking with an LLM confined to prose
- S3 as the sole datastore — write-once staging plus a curated serving tier
- Radar, Watchlist and Briefing screens, plus live counters and a scanning control panel
- Weekly partnership briefing with rationale, sources and next action, exportable as a shareable artifact
- 374 automated tests, 86% coverage, 46 browser tests against live data
- Authenticated API, least-privilege IAM policy, structured logging, resource limits

I.2 Honest limitations

Stated explicitly so a green checklist is not mistaken for total coverage.

Limitation	Status
EPO patent source	Built and unit-tested; disabled pending a free OAuth key
SMTP notifications	Implemented, untested; webhook delivery verified live
AWS deployment	Not done — runs on Docker Compose; infrastructure/ untouched
TLS	The UI↔scanner hop is plaintext inside the compose network
Load evidence	50k signals tested in-process; no data under real S3 throttling
Production burn-in	Days, not months. Ten unattended runs is not yet a track record
Company entity resolution	Improved but imperfect; sibling entities stay separate by design
Commercial sources	Evaluate Pharma and Citeline require paid subscriptions

I.3 Recommended next steps

#	Step	Effort
1	Obtain an EPO OPS OAuth key and enable the fifth source	Minutes once issued
2	Run the supplied IAM provisioning script to move onto a least-privilege credential	One command
3	Configure a Slack webhook and switch notifications on	Configuration only

#	Step	Effort
4	Deploy to AWS — the code already prefers an instance role when no static key is set	Terraform
5	Let the BD team tune scoring weights against real briefings	Ongoing, no code

Assessment

The code is production-grade and the deployment is hardened enough to expose internally. What separates it from a claim of full production readiness is operational maturity — TLS, load evidence under real infrastructure, and time in service — rather than any known defect left unaddressed.

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